Wind-driven resuspension and turbidity recovery in a shallow eutrophic lake

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Abstract

We measured turbidity and wind speed at three stations in Lake Orrin, a shallow eutrophic lake with a mean depth of 2.1 m, during the autumn of 2025. Turbidity rose within two hours of wind speeds exceeding 6 m/s and returned to baseline after a median of 31 hours. A simple first-order decay model explained 78% of the variance in recovery times.

Keywords: sediment resuspension; turbidity; shallow lakes; wind forcing

1 Introduction

Shallow lakes are prone to sediment resuspension because wind-generated waves can reach the bed over most of their area. Resuspended material reduces light availability and can release phosphorus from the sediment.[1] Few studies have followed the full cycle from resuspension to recovery at hourly resolution, and most rely on monthly sampling that misses short events.

In this study we combine continuous sensor records with wind data to describe how quickly the water column clears after a storm, and whether recovery depends on station depth.

2 Methods

2.1 Study site

Lake Orrin covers 4.6 km² and has a maximum depth of 3.4 m. The lake is polymictic and has no thermal stratification lasting more than a few days. Three monitoring buoys were deployed along the main wind axis, from the sheltered western bay to the open centre.

2.2 Measurements

Turbidity was recorded every 10 minutes with optical sensors, and wind speed was measured 4 m above the water surface. Sensors were cleaned and checked against laboratory samples every two weeks.

Table 1. Deployment summary for the three stations.

Station	Depth (m)	Distance from shore (m)	Valid records (%)
S1	1.6	180	97.2
S2	2.4	640	94.8
S3	3.1	1,120	91.5

An event was defined as a rise in turbidity of more than 50% above the seven-day running median, sustained for at least one hour.

[1] Internal loading was not measured directly in this study.

3 Results

Eleven resuspension events were identified between September and November. Peak turbidity was highest at the shallowest station and decreased with depth.

Two events at station S3 were lost to sensor fouling, which explains the lower count there.

Table 2. Recovery times by station (hours).

Station	Events	Median	Range
S1	11	36	18–64
S2	11	31	15–52
S3	9	24	12–41

4 Discussion

Recovery was faster at the deeper stations, where less material was resuspended in the first place. This pattern suggests that the duration of poor light conditions is governed more by the amount of material lifted than by settling distance.

Management measures that reduce fetch, such as restoring reed belts along the western shore, could therefore shorten turbid periods.

5 Conclusions

Hourly monitoring captured the complete resuspension cycle and showed that recovery time scales with station depth.

References

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